

Chapter 23. Mechanical Ventilation in the Neonatal and Pediatric Setting

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Mechanical Ventilation in the Neonatal and Pediatric Setting: Introduction

Respiratory disease in its various forms remains the most common cause of pediatric and neonatal morbidity and mortality. One of the most common reasons for admission to pediatric or neonatal intensive care units is the need for ventilatory support for acute or impending respiratory failure. The major challenge for these units is to deal with a very heterogeneous population of patients who are characterized by enormous differences in age and size and marked developmental changes in organ physiology during growth. In particular, the pediatric intensive care unit population is characterized by a wide variety of rare and unique medical problems that make large clinical trials, even on general topics such as ventilator support, very difficult to conduct.^{1,2} Despite worldwide daily use of mechanical ventilation in pediatric and neonatal intensive care units, many clinical and practical questions remain unresolved. Answers are often extrapolated from the results of adult studies. This may seem sensible for older children but is dangerous when applied to neonates, infants, and children up to the age of 12 years, because of developmental alterations in the physiology of their organ systems, particularly (but not only) their respiratory system.

Peculiarities in Physiology and Respiratory Mechanics

The considerable differences in respiratory physiology and anatomy between infants and adults³ explain why infants and young children have a higher susceptibility to more severe manifestations of respiratory diseases, and why respiratory failure is a common problem in neonatal and pediatric intensive care units ([Table 23-1](#)). The appreciation of the peculiarities of pediatric respiratory physiology is essential for correct management of critically ill and/or ventilated infants and children.

Table 23-1: Physiologic Reasons for the Increased Susceptibility for Respiratory Compromise of Infants in Comparison to Adults

Cause	Physiologic or Anatomic Basis
Metabolism ↑	↑ O ₂ consumption
Risk for apnea ↑	Immaturity of control of breathing
Resistance to breathing ↑	
Upper airway resistance ↑	Nose breathing
	Large tongue
	Airway size ↓
	Collapsibility ↑
	Pharyngeal muscle tone ↓
	Compliance of upper airway structures ↑
Lower airway resistance ↑	Airway size ↓
	Collapsibility ↑
	Airway wall compliance ↑
	Elastic recoil ↓
Lung volume ↓	Numbers of alveoli ↓
	Lack of collateral ventilation
Efficiency of respiratory muscles ↓	Efficiency of diaphragm ↓
	Rib cage compliance ↑
	Horizontal insertion at the rib cage
	Efficiency of intercostal muscles ↓
	Horizontal ribs
Endurance of respiratory muscles ↓	Respiratory rate ↑
	Fatigue-resistant type I muscle fibers ↓

From Hammer J, Eber E. The peculiarities of infant respiratory physiology. In: Hammer J, Eber E, eds. *Paediatric Pulmonary Function Testing. Prog Respir Res*. Basel, Switzerland: Karger; 2005;33:2–7.

The basal metabolic rate is approximately twofold higher in infants than in adults (7 mL/kg/min at birth vs. 3 to 4 mL/kg/min in the adult). Hence, the normal resting state in infants is already one of high respiratory and cardiovascular activity. This means that infants have less metabolic reserve if O₂ consumption needs to be increased during critical illnesses.

Control of Breathing

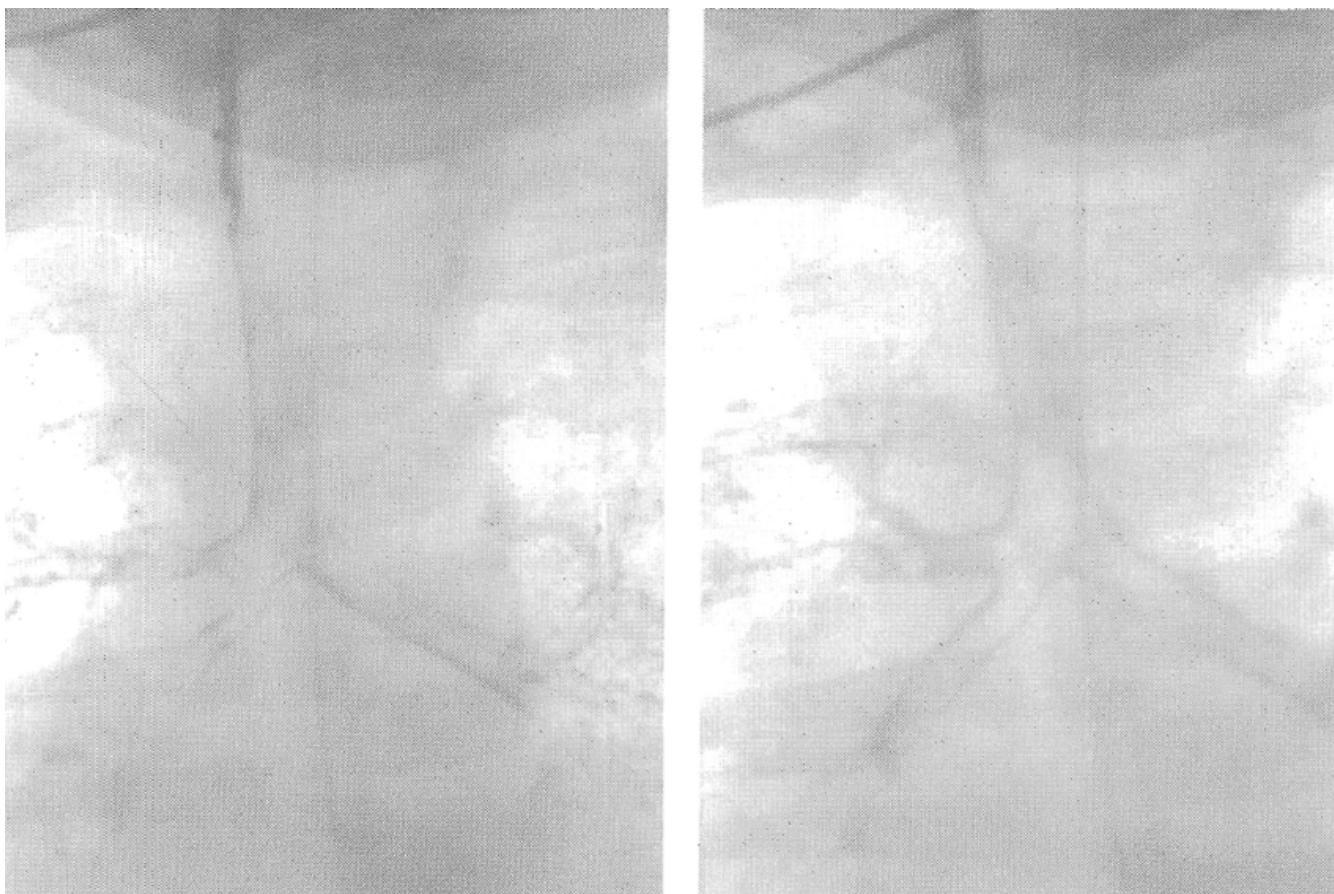
A considerable amount of maturation of the control of breathing occurs in the last few weeks of gestation and in the first few days of life, which explains the high prevalence of apnea in infants born prematurely.⁴ The breathing pattern of newborn, especially premature, infants is irregular with substantial breath-to-breath variability and periodic breathing at times, which increases the risk of prolonged, potentially life-threatening apnea under certain circumstances. The responses to hypercapnia or hypoxia are decreased and of variable sensitivity, making the young infant much more vulnerable to any noxious stimuli and disturbances of the respiratory control mechanisms.⁵

Ineffective breathing can lead to hypoxemia and bradycardia that may be severe enough to require the use of continuous positive airway pressure (CPAP) or intubation and mechanical ventilation, especially in the very preterm baby. Methylxanthines (such as caffeine) are effective in reducing the need of ventilator support for apnea of prematurity.⁴ Feedback from vagal stretch reflexes, rib cage muscles, and the changing mechanical state of the respiratory system during each breath influence respiratory activity. If inflation is small, there is little inhibitory vagal feedback. If inflation is excessive, inspiration is inhibited. This inspiratory-inhibitory reflex, the *Hering-Breuer inflation reflex*, is very potent in the preterm and less so in the term newborn and young infant during the first weeks of life.⁶ This inflation reflex facilitates passive measurement of respiratory mechanics, but may interfere with ventilator triggering. Babies may become apneic after a mechanical inflation, especially if an excessively prolonged inspiratory time (T_I), is used.⁷ Fortunately, CPAP seems to enhance a baby's ability to adjust to increased respiratory loads, possibly by the elimination of the Hering-Breuer deflation reflex.⁸

Upper and Lower Airways

In the newborn, nasal breathing is obligatory, or at least strongly preferential, secondary to the configuration of the upper airways. The epiglottis is relatively large, floppy, and positioned high in the pharynx so that it is in contact with the soft palate, thereby favoring nasal over mouth breathing.⁹ The larynx, trachea, and bronchi are considerably more compliant than in the older child, thus making the infant's airway highly susceptible to distending and compressive forces.¹⁰ Thus, with any obstruction of the upper airway, significant dynamic inspiratory collapse of upper airway structures can occur during forceful inspirations, which further adds to the obstruction already present. With lower-airway obstruction, forced expiratory efforts result in increased intrathoracic pressure and dynamic expiratory lower airway collapse further limiting expiratory flow. Positive end-expiratory pressure (PEEP) may help to stent collapsing airways during expiration in some patients (e.g., those with tracheobronchomalacia; Fig. 23-1).¹¹

Figure 23-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Bronchography in patient with tracheobronchomalacia without application of positive airway pressure (*left*) and with 10 cm H₂O of positive airway pressure (*right*). Application of positive airway pressure produces stenting of the trachea and the bronchial tree. (Courtesy Quen Mok, Great Ormond Street Hospital for Children, London, UK.)

Small peripheral airways contribute approximately 50% to the total airway resistance in the infant lung (compared to approximately 20% in the adult). As airway diameter and length increase with age and growth, airway resistance falls tremendously from birth to adulthood.¹² Therefore, diseases that affect the small airways and cause large changes in peripheral resistance may be clinically silent in an adult, but can cause significant problems in infants (e.g., bronchiolitis).

The airways of a child are relatively large in comparison with those of an adult, although in absolute terms they are small. Airway resistance in spontaneously breathing infants is normally 20–30 cm H₂O/liter per second; values in intubated infants are 50 to 150 cm H₂O/L/s, consequent to the diameter of the endotracheal tube (ETT).¹³ For many years it was thought that small infants could not breathe spontaneously through an appropriately sized ETT (because of the high resistance imposed by the small lumen), but this view is no longer maintained. Under normal tidal flow conditions, resistance imposed by a small tube is not substantially higher than that with larger tubes.¹⁴

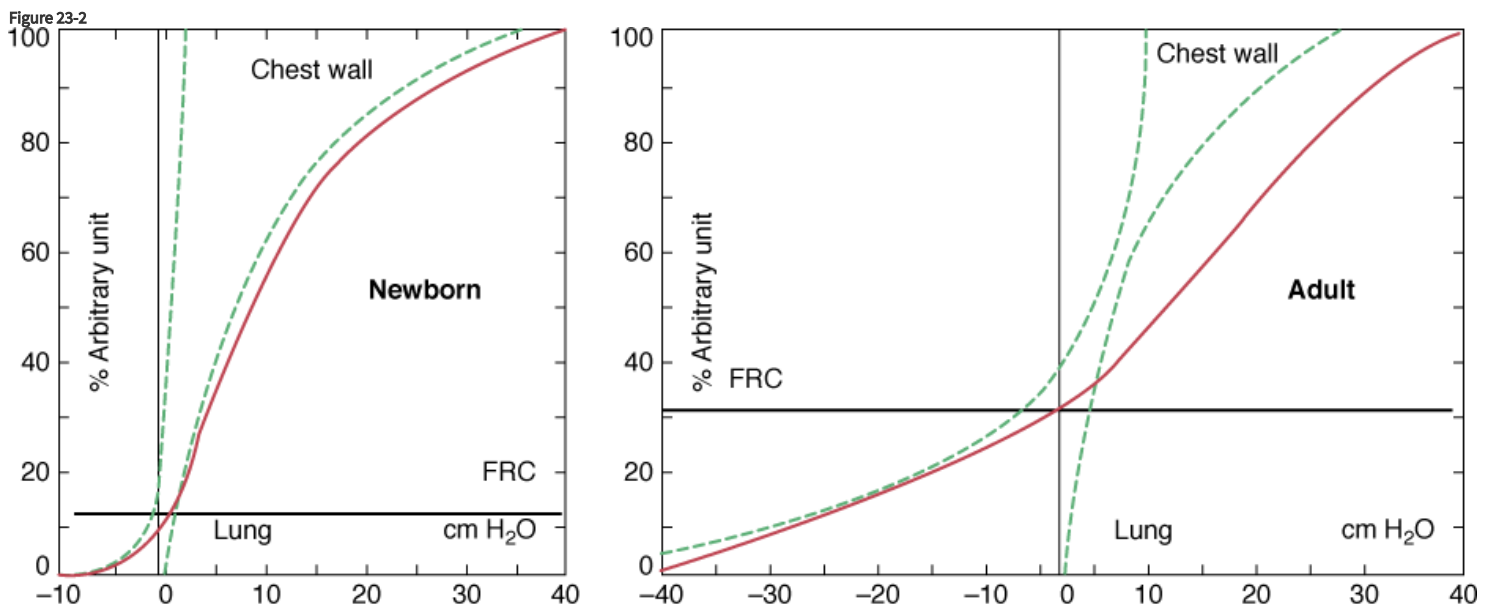
Lung and Chest Wall

Because of its shape, high compliance, and deformability, the contribution of the rib cage to tidal breathing is limited in newborns and infants. The ribs are horizontally aligned and allow for less anteroposterior movement of the chest wall and less efficiency of the intercostal muscles during respiration. The highly compliant chest wall is easily distorted, so that under conditions of respiratory impairment, much energy is wasted by sucking in ribs rather than fresh air. This paradox inward movement of the chest wall during inspiration is a common sign of almost any disorder causing respiratory distress in infants, but is most pronounced in upper-airway obstruction.

The elastic tissue in the septa of the alveoli surrounding the conducting airways provides the elastic recoil that enables the airways to remain open. Early in life there are few relatively large alveoli that provide little support for the airways, which are thus able to collapse

easily. Addition of alveoli continues throughout early childhood by septal division, providing more elastic recoil and a decreased tendency for airway collapse with increasing age.¹⁵ Elastic recoil increases until adolescence and declines again with aging. Collateral pathways of ventilation (intraalveolar pores of Kohn and bronchoalveolar canals of Lambert) do not appear until 3 to 4 years of age,^{16,17} which excludes alveoli beyond obstructed airways to be ventilated by these alternate routes and predisposes the infant to the development of atelectasis.

The balance between the chest and lung recoil pressure that determines the static resting volume of the lung is set at lower lung volumes in infants compared with older children (Fig. 23-2).¹⁸ The highly compliant chest wall results in relatively low transpulmonary pressures at end-expiration that leads to an increased tendency for collapse of the small peripheral airways and ventilation inhomogeneity. The infant adopts several strategies to constantly establish lung volumes, including (a) a higher respiratory rate with insufficient time to exhale to the elastic equilibrium volume, (b) laryngeal adduction during exhalation to increase the resistance to airflow, and (c) constant postinspiratory diaphragmatic muscle activity (i.e., tonic diaphragmatic activity). A PEEP of 5 to 6 cm H₂O is necessary to maintain end-expiratory lung volume in infants under neuromuscular blockade.¹⁹ The highly compliant chest wall also explains the generally high cardiovascular tolerance of infants to high airway pressures that are less efficiently transmitted to the pleural space. The low elastic recoil of the chest, together with a low percentage of slow-muscle fibers in both the diaphragm and intercostal muscles, explain why very little airway pressure is needed to expand the chest wall during inspiration in healthy babies.



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Characteristics of lung and chest wall mechanics in newborns compared to adults. The compliant chest wall in the newborn leads to lower functional residual capacity (FRC) in the newborn. (Based on data published in Agostoni E. Volume-pressure relationships of the thorax and lung in the newborn. *J Appl Physiol*. 1959;14:909-913.)

Circulatory Changes Pertinent to Mechanical Ventilation after Birth

Before birth, only approximately 10% of blood flow passes through the lungs as a result of high pulmonary vascular resistance, patent ductus arteriosus, open foramen ovale, and the low-resistance placental component to the systemic circulation. Pulmonary vascular resistance falls rapidly in the first minutes of life, but is still elevated and falls only gradually to normal levels over days to weeks. Under certain circumstances (e.g., birth asphyxia, chronic hypoxia, or neonatal septicemia with or without metabolic acidosis), pulmonary vascular resistance increases or remains high, leading to right-to-left shunting through the ductus arteriosus and sometimes even through the foramen ovale. This persistent pulmonary hypertension of the newborn (PPHN) leads to poor systemic oxygenation, as revealed by low postductal transcutaneous saturation levels or desaturation of all four extremities when the right-to-left shunt at the level of the foramen ovale is important.

High inflation pressures to recruit lung volume and improve oxygenation should be used with caution because of the risk of increased pulmonary vascular resistance by overdistension and compromised cardiac output. First-line treatment is directing toward known determinants of pulmonary vascular resistance (Table 23-2) such as acidosis or high interstitial pressures. Second-line therapy includes specific pulmonary vasodilators, such as inhaled nitric oxide (iNO)^{20–22} after excluding congenital heart disease with duct-dependent systemic circulation, which may deteriorate when pulmonary vascular resistance is lowered (e.g., total anomalous pulmonary venous return or hypoplastic left-heart syndrome).²³

Table 23-2: Main Determinants of Pulmonary Vascular Resistance

Increase in Pulmonary Vascular Resistance	Decrease in Pulmonary Vascular Resistance
High interstitial pressure	Low interstitial pressure
High lung volumes (alveolar overdistension)	Normal lung volumes
Low lung volumes (atelectasis)	
Low alveolar P _{O2}	High alveolar P _{O2}
Low arterial pH	High arterial pH

P_{O2}, partial pressure of oxygen.

Oxygen Transport

Fetal hemoglobin is the predominant type of hemoglobin at birth and decreases steadily over the first 6 months. 2,3-Diphosphoglycerate (2,3-DPG) has a lower affinity to fetal hemoglobin and shifts the dissociation curve to the left. This facilitates loading and unloading of O₂, ensuring, together with the high O₂-carrying capacity of fetal hemoglobin, adequate tissue oxygenation despite low partial pressure of arterial oxygen (Pa_{O2}) values in utero (15 to 30 mm Hg). The predominance of fetal hemoglobin makes it possible, if necessary, to tolerate lower Pa_{O2} values (but not really arterial oxygen saturation [Sa_{O2}]) better in early postnatal life than is possible later in life.

Preterm babies can maintain their growth in utero with a mean Pa_{O2} of 3.2 kilopascals (kPa; approximately 24 mm Hg), equivalent to Sa_{O2} of approximately 70%. In neonatal intensive care units, clinical practice has long been to keep oxygen levels in preterm newborns in line with those of term infants until neonatal morbidities induced by oxygen have been better understood. To date, there is insufficient evidence to suggest what would be the optimal Sa_{O2} or Pa_{O2} values in preterm infants to avoid potential oxygen toxicity while ensuring adequate oxygen delivery to tissues. There is, however, ample evidence to support the notion that a “restrictive” oxygen approach does more good than harm in preterm babies.²⁴

Some Special Considerations for Ventilator Management in Neonates, Infants, and Children

The basic objectives of ventilator support remain the same from early life to adulthood^{25,26} but ventilator strategies, modes, and targets are not always fully identical to those established in adults.

Lung-Protective Ventilation

Barotrauma zones, safe peak inspiratory pressure and PEEP, are less-well defined in the younger age group, especially the very premature infant. Neonatologists tend to limit peak inspiratory pressure to 30 cm H₂O or lower; some favor the use of high-frequency oscillation ventilation (HFOV) in this age group. Studies investigating lung-protective ventilation in neonates have mainly focused on comparing high-

frequency ventilation with conventional mechanical ventilation (CMV), leaving unanswered the important question as to whether reducing tidal volume or increasing PEEP is also lung protective.²⁷ HFOV is best used with a high-lung-volume strategy, but offers no benefit over conventional ventilation when a lung-protective strategy is applied.²⁸

A lung-protective strategy is commonly advocated similar to recommendations for adults following the publication of the ARDS Network trial.²⁹ Recent observational cross-sectional studies, however, revealed that commonly used tidal volume (V_T) varies between 6 and 10 mL/kg (mean: 8.3 ± 3.3 mL/kg) in children,^{2,30} and between 4 and 6 mL/kg (mean: 5.7 ± 2.3 mL/kg) in neonates.³¹ Interestingly, V_T up to 10 mL/kg in children does not seem to be associated with increased mortality. Moreover, higher V_T within this range is associated with more ventilator-free days, particularly in patients with less-severe disease.³⁰ In a recent prospective, multicenter, observational study, higher maximum and median tidal volumes were associated with reduced mortality, even when corrected for severity of lung disease.³² There is even less data available for neonates than for children to confirm or reject the hypothesis that small-volume ventilation, V_T of 4 to 6 mL/kg, is the best lung-protective approach.²⁷

Allowance for Hypercapnia or Hypocapnia, Respectively

Overventilation and hypocapnia have been shown to contribute to adverse neurodevelopment in preterm infants and also in the neonate with hypoxic ischemic encephalopathy.^{33–35} Any rapid reduction in partial pressure of arterial carbon dioxide (P_{aCO_2}), which may occur with surfactant therapy or change in ventilator mode, may lead to cerebral vasoconstriction, increasing the risk of cerebral hemorrhage and/or leukomalacia. Both extremes and fluctuations of P_{aCO_2} are associated with severe intraventricular hemorrhage. It may be prudent to avoid extreme hypocapnia and hypercapnia during the period of risk for intraventricular hemorrhage in preterm infants. Both minimum partial pressure of carbon dioxide (P_{CO_2}) and cumulative P_{CO_2} less than 35 mm Hg are associated with poor outcome in neonates with hypoxic ischemic encephalopathy.³⁴

Volume-Targeted Ventilation in Neonates

Modern neonatal ventilator modes can target a set tidal volume as an alternative to traditional pressure-limited ventilation. Volume-targeted ventilation (VTV) aims to produce a more stable V_T so as to reduce lung damage and stabilize P_{aCO_2} . Infants receiving VTV had lower rates of death and chronic lung disease compared with infants ventilated using pressure-limited ventilation modes.^{36,37} Further studies are needed to identify whether VTV modes improve neurodevelopmental outcomes and to compare and refine VTV strategies.

VTV has some technical limitations. First, the displayed VT can be misleading in the presence of a high ETT leak. Such leak is common in neonatal practice where uncuffed tubes are commonly used and may be noted in approximately 75% of all ventilated neonates, usually without greater clinical relevance.³⁸ Leaks are associated with an inability to detect the patient's inspiratory effort or autotriggering of the ventilator. The latter also occurs if tubing is not kept dry and if there is water accumulation in the expiratory limb. This complication is more common in the neonate and young infant. Second, inaccurate V_T delivery under specific conditions, especially when rapid changes in respiratory system mechanics occur, has been described with various neonatal ventilators that offer VTV. Such discrepancies between the set V_T and the delivered inflations can be harmful in clinical situations, especially in newborns. Their clinical relevance needs to be clarified with safety studies in the neonatal population, before the “uncritical” use of such new hybrid modes could be advocated for all neonates and for various clinical conditions.³⁹

Setting Ventilator Parameters According Respiratory Mechanics

There is a tendency towards using very short inspiratory times in premature infants with respiratory distress syndrome (RDS) secondary to the very low compliance of the lungs. An increase in inspiratory time may sometimes result in lower peak inspiratory pressures and more homogenous ventilation. A good knowledge of respiratory mechanics is helpful to decide on the best settings. In addition, there exists the erroneous belief that expiratory times have to be longer than inspiratory times. While this is the case in obstructive airways disease, it does not apply to the healthy or the stiff lung in neonates and children.

Small Endotracheal Tubes

The small and narrow ETTs are prone to mucus plugging and kinking. Many clinicians believe that for an infant or young child, breathing through a small ETT is equivalent to breathing through a straw, thereby imposing an unacceptable work of breathing. This notion is contrary to both clinical observation and physiology.^{14,40} A 3-kg infant accepts a 3-mm inner-diameter ETT, whereas a 60-kg adult can tolerate a 9-mm inner-diameter ETT—a 20-fold increase in body size but only a threefold increase in ETT size. The narrower tube is irrelevant because it is shorter and because lower flows are generated by the infant compared to the adult. The net effect is that the infant is breathing through a hose rather than a straw when compared to the adult. Hence, if an infant or young child cannot sustain a spontaneous breathing trial on CPAP or a T-piece for several hours, he or she is as likely to fail extubation as when pressure support is applied. Furthermore, the addition of pressure support is likely to mask respiratory insufficiency and contribute to a higher failed extubation rate.⁴¹

Tracheotomy for Long-Term Ventilation

Tracheotomy is withheld for a longer period in critically ill children than in adults. Infants tolerate prolonged intubation much better than adults, although complications, such as subglottic stenosis,⁴² can occur. Furthermore, indications for tracheotomy are different between children and adults: Tracheotomy is more often appropriate for bypassing congenital or acquired upper-airway obstruction than for supporting long-term mechanical ventilation.⁴³

Closed Suction Systems

There is probably no medical device other than closed-suction systems that has been as enthusiastically accepted in intensive care unit practice without the requirement to prove benefit in a randomized controlled trial. There is still insufficient evidence to decide between endotracheal suctioning with or without disconnection.⁴⁴ In the injured lung, however, closed suction is much less effective than open suction, irrespective of the type of secretions.⁴⁵ The negative effects of suction on lung volume, heart rate, and SaO_2 are transient and highly variable with both open-suction and closed-suction methods in spontaneously breathing infants.⁴⁶ This suggests that it is time to reassess the role of open suction, especially in ventilated infants with narrow tubes and respiratory disease resulting in large amounts of copious secretions.

Common Clinical Conditions

Neonates

Respiratory Distress Syndrome (Hyaline Membrane Disease)

RDS is characterized by pulmonary surfactant deficiency and transudation of plasma proteins into the alveolar spaces. It typically occurs in preterm infants, and its severity correlates with the degree of immaturity of the lungs and is aggravated by concomitant infection, oligohydramnios, and a variety of other factors. The results are stiff lungs and a tendency to atelectasis. Initial ventilator management aims to recruit collapsed alveoli, restore functional residual capacity, and achieve adequate alveolar ventilation. Technical and pharmacologic advances, with improved understanding of pathophysiology and causes of lung injury, have significantly enhanced the ventilator management of preterm infants with RDS. Ventilator modalities have shifted towards early noninvasive ventilation and decreased and shortened use of invasive forms of ventilation in most preterm infants.

Nasal CPAP applied by various devices has become the primary respiratory support for RDS. When used early, it reduces respiratory failure, the need for surfactant, and the duration and invasiveness of respiratory support without impairing neonatal outcome.⁴⁷ It therefore offers a valuable alternative to intubation and surfactant in preterm infants.⁴⁸ It has not, however, proven to allow for better outcome than intubation and mechanical ventilation in preterm infants.⁴⁹ Nasal CPAP improves respiration in preterm infants by increasing functional residual capacity and chest wall stability, as well as decreasing upper-airway collapsibility and upper-airway resistance.⁵⁰

Recently, heater-humidifier devices that use novel methods for conditioning respiratory gases from an external source have been introduced. The addition of sufficient warmth and high levels of humidification to a gas has enabled the use of higher flow rates from a

nasal cannula. High-flow nasal cannula can be used to provide high concentrations of O_2 and may deliver PEEP. At present, there is insufficient evidence to establish the safety or effectiveness of high-flow nasal cannula as a form of respiratory support in preterm infants. When used following extubation, high-flow nasal cannula seems to result in a higher rate of reintubation than does nasal CPAP.⁵¹

When positive-pressure ventilation was first introduced in newborn infants with RDS, the use of high peak inspiratory pressures was associated with high mortality rates, air leaks, and the development of bronchopulmonary dysplasia (BPD).⁵² In the 1970s, the incidence of pneumothoraces was reduced by adopting a strategy using long T_I values and slower rates.⁵³ This strategy was widely adopted and only given up after the benefit of PEEP was better understood.⁵⁴ Administration of exogenous surfactant and advancements in ventilator technology, which decrease patient-ventilator asynchrony, have further improved the management and outcome of RDS. It is now widely accepted that severe RDS, characterized by poorly compliant lungs and very short time constants, is best managed with short T_I values (0.26 to 0.34 seconds), rapid rates (60/breaths/min or more), and PEEP.^{55,56} Such general recommendations help in avoiding major errors when initiating mechanical ventilation in infants suffering from severe RDS. Nevertheless, inspiratory and expiratory time, respiratory frequency, and PEEP need to be customized to an individualized assessment of a patient's respiratory mechanics, which change over the course of disease or with therapeutic interventions such as surfactant administration. The availability of low-dead-space-flow sensors enables continuous measurement of V_T , and eliminated much of the guesswork in setting the ventilator. Volume-targeted modes have now become increasingly popular in the neonatal intensive care unit to improve the stability of ventilation and to reduce unnecessarily high peak pressures.³⁷ This strategy aims at reducing Pa_{CO_2} fluctuations in very preterm infants (although it is not always as successful as hoped^{57,58}), which are associated with the development of intraventricular hemorrhage and leukomalacia. Hence, much emphasis is placed on using ventilator strategies that should best guarantee a constant Pa_{CO_2} and at the same time use of the lowest fractional inspired oxygen concentration (FI_{O_2}) to achieve sufficient tissue oxygenation.

The underlying pathophysiology, decreased lung compliance, high chest wall compliance, and dynamically maintained functional residual capacity above closing volume make RDS a perfect candidate for an open-lung strategy and use of HFOV. Despite many studies of HFOV versus conventional ventilation (and three meta-analyses of these trials) involving 3652 infants, neonatologists remain unsure about the potential benefits and harms of HFOV for support of preterm infants with surfactant-deficient lungs. The Prevention of Ventilator Induced Lung Injury Collaborative Group (PreVILIG collaboration) recently concluded that HFOV seems equally effective to conventional ventilation in preterm infants.⁵⁹ Current results do not support selection of preterm infants for HFOV on the basis of gestational age, birthweight for gestation, initial lung disease severity, or exposure to antenatal corticosteroids.⁵⁹ Why data from animal models predicted a larger effect size than that observed in human infants remains an unanswered question. Recent data, however, support the usefulness of first-intention HFOV strategy in managing selected subpopulations, such as very-low-birthweight newborns complicated by severe RDS not antenatally treated with glucocorticoids.⁶⁰

Exogenous surfactant therapy has been part of routine care of preterm neonates with RDS since the early 1990s. Animal-derived exogenous surfactants are the present treatment of choice, with few adverse effects. The few adverse effects are largely related to changes in oxygenation and heart rate during surfactant administration.⁶¹ The optimal dose is usually 100 mg/kg. Both prophylaxis and early treatment are successful in infants with established RDS, but prophylaxis appears to produce greater benefit.⁶² The concept of using surfactant early in patients with low oxygen needs ($FI_{O_2} < 0.45$), combined with the so-called INSURE approach (intubation-surfactant-extubation to nasal CPAP), has decreased the need for mechanical ventilation, incidence of BPD, and air leak syndromes as compared to subsequent use of INSURE when FI_{O_2} exceeds 0.45.^{47,63}

Persistent fetal circulation is a complication of RDS in preterm infants. At present, there is no clearcut evidence in favor of using iNO for preterm infants requiring mechanical ventilation.⁶⁴ Clinicians continue to make case-by-case decisions for the treatment of preterm infants with hypoxia unresponsive to other therapies.⁶⁵

Persistent Pulmonary Hypertension of the Newborn

PPHN is the result of elevated pulmonary vascular resistance to the point that venous blood is shunted to some degree through fetal channels (e.g., ductus arteriosus and foramen ovale) into the systemic circulation. By bypassing the lungs, it causes systemic arterial hypoxemia. It is associated with (a) pulmonary parenchymal disease such as RDS or meconium aspiration and (b) hypoplasia of the lungs, most often in the form of diaphragmatic hernia. It (c) also can occur without an evident cause.

Traditionally, these infants have been hyperventilated to achieve mild hypocapnic alkalosis in an attempt to attenuate hypoxic pulmonary vasoconstriction. iNO is an approved adjunct to improve oxygenation in term and preterm infants with severe hypoxemic respiratory failure secondary to PPHN and has been shown to reduce the need for extracorporeal membrane oxygenation.⁶⁶ Alveolar recruitment should be optimized to achieve best iNO effects on oxygenation.²³ Hence, HFOV is an ideal method to combine with iNO.

Meconium Aspiration Syndrome

Meconium aspiration syndrome affects mature infants and is characterized by a mixture of inflammatory pulmonary disease, secondary surfactant deficiency, obstruction of small airways, and pulmonary hypertension. The result is a very inhomogeneous lung with areas of atelectasis and hyperinflation. In terms of pathophysiology and ventilator management, meconium aspiration syndrome resembles acute respiratory distress syndrome (ARDS) of pulmonary origin, favoring the use of similar ventilator strategies (open lung concept, permissive hypercapnia, and so on). Despite the introduction of innovative ventilatory treatments for this disease over the last two decades (e.g., surfactant-lavage, HFOV, iNO, extracorporeal membrane oxygenation), most infants can be successfully managed with CPAP or mechanical ventilation alone.^{67,68}

Congenital Diaphragmatic Hernia

Congenital diaphragmatic hernia (CDH) is characterized by lung hypoplasia, surfactant deficiency, and an extremely reactive hypoplastic pulmonary vascular system. The hypoplastic lungs are very vulnerable to injury from aggressive ventilation with high inspiratory pressures. Treatment is no longer focused on immediate surgery, but rather on delaying repair until all reversible problems are resolved (i.e., stabilization period).^{69,70} The concept of delaying repair until pulmonary hypertension resolves and the patient is hemodynamically stable has increased survival from below 50% to as high as 80%.^{69,71} Although high airway pressures may be required with secondary parenchymal injury to ensure adequate oxygenation, special care is necessary to avoid alveolar distension in the setting of pulmonary hypoplasia. Neither the use of HFOV nor early rescue extracorporeal membrane oxygenation, however, has significantly improved outcome in CDH patients, the best survival rates with these options are approximately 80%.⁷²⁻⁷⁷ The discussion on optimal treatment strategies for CDH, in addition to the commonly accepted need for early lung-protective ventilation to avoid lung injury, is not closed.⁷⁸

Pediatric Patients

Virus-Induced Hypoxemic Failure

Two distinct patterns of disease occur in infants with virus-induced respiratory failure.^{79,80} The primary causative agent is respiratory syncytial virus. The most frequent pattern is acute bronchiolitis characterized by an obstruction of small airways with air trapping and a moderate degree of parenchymal disease from atelectasis. This leads to increased respiratory resistance, a prolonged time constant, intrinsic PEEP, and decreased respiratory compliance.⁸¹ Radiographic findings include hyperinflation, perihilar infiltrates, and atelectasis. The second pattern, which affects approximately 25% to 30% of infants with respiratory syncytial virus-induced respiratory failure, consists of severe restrictive parenchymal disease (usually termed *respiratory syncytial virus-pneumonia*). Typically, respiratory compliance and lung volumes are decreased markedly without significant airway obstruction or air trapping. Alveolar consolidation is the main radiographic feature. This subgroup also usually fulfills the criteria of ARDS. Patients require prolonged ventilation compared with a brief duration (4 to 8 days) for the bronchiolitic pattern.^{79,80}

No consensus exists on optimal ventilator strategy. Nasal CPAP appears useful in the early stages of severe bronchiolitis by decreasing the work of breathing needed to overcome the intrinsic PEEP. The evidence supporting the use of CPAP to reduce P_{CO_2} and respiratory distress in bronchiolitis, however, is of low methodologic quality, reporting only short-term effects,⁸² and there is no conclusive evidence that CPAP

reduces the need for intubation.^{83,84} More recently, high-flow nasal cannula, if used early in the treatment of severe bronchiolitis, appears to reduce the need for intubation in infants with viral bronchiolitis.⁸⁵

Pressure-controlled ventilation (PCV) is the mode used most commonly in virus-induced respiratory failure because the decelerating flow pattern achieves a lower mean airway pressure than volume-controlled ventilation (VCV). Most patients need peak inspiratory pressures of 25 to 35 cm H₂O to achieve adequate ventilation. Infants with obstructive disease have long expiratory time constants. They are best ventilated with slow rates and inspiratory-to-expiratory (I:E) ratios of at least 1:3 to prevent breath “stacking” and further hyperinflation. Conversely, patients with restrictive disease may require faster rates and lower I:E ratios. Permissive hypercapnia, to avoid high peak inspiratory pressures and barotrauma, can be used with both pathophysiologic patterns.

Obstructive patients may do poorly with HFOV, although this statement is not as absolute as often thought.⁸⁶ Patients with predominant restrictive disease, however, are better candidates for HFOV.

Pediatric Acute Respiratory Distress Syndrome

The causes of ARDS in pediatric patients are the same as in adults, although viral or bacterial respiratory infections are more common in children.⁷⁹ Ventilator strategies are the same as in adult patients and include high PEEP and limiting peak inspiratory pressure to below 30(–35) cm H₂O. Lung-protective ventilation strategies are commonly recommended, although there is doubt as to the best V_T.

Although no single ventilation mode has proven superior, the most common modes are PCV and HFOV,^{2,30,87} despite the absence of solid data on the use of HFOV in pediatric patients beyond the neonatal period. Prone positioning is used commonly, but without proven benefit.^{88,89} iNO cannot be recommended for routine use, although it is used as rescue therapy (for a short time) in the very early stages of life-threatening hypoxemia and severe pulmonary hypertension.

Status Asthmaticus

Most complications in patients with asthma receiving ventilation occur during or immediately after intubation. They result largely from gas trapping that causes hypotension, O₂ desaturation, pneumothorax, and cardiac arrest. Institution of positive-pressure ventilation in patients with asthma alters cardiocirculatory and respiratory dynamics dramatically, leading to diminished venous return and hypotension.

The optimal ventilator mode for status asthmaticus is not established. Randomized, controlled trials comparing different modes in pediatric asthma patients are virtually impossible.⁹⁰ Most clinicians prefer pressure-limited ventilation, keeping peak inspiratory pressure below 35 to 40 cm H₂O and accepting hypercapnia.⁹¹ Because of their decelerating flow pattern, PCV and pressure-regulated VCV allow for lower peak inspiratory pressures but result in higher mean airway pressures than VCV with identical V_T. Ventilator rates are set well below normal and require an extremely long expiratory and reasonably long inspiratory time. Externally applied PEEP should be, in general, just below auto-PEEP to decrease trigger work but not increase hyperinflation. Duration of mechanical ventilation for status asthmaticus is usually of 1 to 4 days. Children with rapid-onset near-fatal asthma may have a shorter duration of ventilation than children with asthma that progresses slowly to respiratory failure.⁹²

HFOV, pressure-support ventilation (PSV), and noninvasive ventilation have been tried in status asthmaticus. HFOV is believed to be contraindicated in children with severe airflow obstruction, although recent experience challenges this belief.⁹³ PSV enables active exhalation, which may decrease hyperinflation. PSV allows patients to determine their own respiratory pattern (rate, T_I, and V_T) and may decrease patient-ventilator dyssynchrony. PSV decreases work of breathing by partially unloading the respiratory muscles. Careful selection of PEEP decreases the work of triggering during PSV. PSV of 22 to 37 cm H₂O has been used successfully in children with asthma requiring full or near-full support, resulting in rapid improvement in gas exchange.⁹⁴ Noninvasive PSV is often poorly tolerated without sedation, and its role in avoiding intubation in children with status asthmaticus remains unclear.^{95,96}

Correct Timing of Extubation

Avoidance of unnecessary weaning delays is important for reducing the risk of nosocomial infection, ETT damage to the airway, development of chronic lung disease in neonates, and prolonged dependency on narcotic drugs.^{97,98} Aggressiveness with early weaning, however, must be balanced against the risk of extubation failure that occurs in 2% to 20% of children^{41,99} and up to 40% of neonates,¹⁰⁰ and which is associated with increased mortality.¹⁰¹

Extubation Readiness

Several indices have been developed to predict successful weaning and extubation. Although these indices have been used in research studies, they have not found widespread use in clinical care, likely because of their complexity and lack of proven benefit over clinical judgment. A trial of spontaneous breathing with a T-piece or the use of 10 cm H₂O of PSV for up to 2 hours has been successful in predicting extubation failure in children.¹⁰² A modified rapid shallow breathing index, also known as frequency-to-tidal-volume ratio, and the compliance, rate, oxygenation, and pressure index has been proposed for weaning prediction in children.¹⁰³

There is fertile ground for future research to identify simple predictors of extubation readiness that would likely shorten the duration of ventilation, decrease length of intensive care unit stay, and potentially reduce ventilator-induced lung injury.⁴¹ Table 23-3 lists proposed clinical and laboratory criteria for weaning failure during 2 hours on CPAP of 5 cm H₂O or a T-piece. The single most common cause of extubation failure in children, however, is upper-airway obstruction, which is difficult to predict with an ETT in place. The air-leak test (an audible air leak when airway pressure ≥25 cm H₂O), although often recommended, has repeatedly been shown to have low sensitivity for testing extubation readiness.^{104–106}

Table 23-3: Criteria for Extubation Readiness Test Failure

Proposed criteria for failure of extubation readiness during 2 hours on CPAP <5 cm H ₂ O or T-piece
Clinical Criteria: Diaphoresis Nasal flaring Increasing respiratory effort Tachycardia (increase in heart rate >40 bpm [breaths per minute]) Cardiac arrhythmias Hypotension Apnea
Laboratory Criteria: Increase in end-tidal CO ₂ >10 mm Hg Decrease of arterial pH <7.32 Decline in arterial pH >0.07 PaO ₂ <60 mm Hg with an FI _{O2} >40 (P/F O ₂ ratio <150) SpO ₂ declines to <5%

From Newth, et al. Weaning and extubation readiness in pediatric patients. *Pediatr Crit Care Med*. 2009;10:1–11.

Extubation failure is more frequent in children with chronic respiratory disorders, neuromuscular and chronic neurologic disorders, and upper-airway problems.¹⁰¹ The most common cause of extubation failure in preterm infants is apnea with bradycardia.¹⁰⁷

Developing reliable thresholds for indices for weaning extremely low-birthweight preterm infants has proven equally difficult as for older children.¹⁰⁸ The percentage of time spent below a target value of spontaneous expiratory minute ventilation (<125 mL/min/kg) during a 2-

hour CPAP trial appears promising but requires further testing.^{109,110} Another option might be a spontaneous breathing trial of 3 minutes on CPAP after minimal ventilator settings have been reached.¹¹¹

Weaning Modes and Adjuncts to Weaning

The most common approach to weaning infants and children is gradual reduction of ventilator support by decreasing rate and airway pressures during synchronized intermittent mandatory ventilation (SIMV). Other weaning methods include PSV and volume-support ventilation. Both methods involve patient-triggered pressure support. With PSV, the level of PSV is adjusted gradually by the physician to a minimal level that achieves an acceptable breathing pattern. With volume-support ventilation, the level of PSV is adjusted continuously by the ventilator to achieve minimum minute ventilation. It is common practice to extubate infants and children from a low level of ventilator support. Gradual weaning is often unnecessary after the underlying disorder has resolved.^{102,112}

Today, many modes have been developed to facilitate weaning by reducing work imposed by the respiratory apparatus.^{113,114} Misperceptions about the impact of the ETT resistance may interfere with weaning management. Although it has become fashionable to use pressure support with PEEP, rather than CPAP or T-piece breathing, to overcome ETT resistance, physiologic and scientific evidence shows that the increase in resistance caused by an appropriately sized tube is minimal and the additional work of breathing negligible. Hence, an infant or young child should be able to sustain a spontaneous breathing trial on CPAP or a T-piece for several hours, if he or she is likely to be successfully extubated. A spontaneous breathing trial using pressure support set at higher levels for smaller ETTs overestimates readiness for extubation in children and contributes to a higher failed extubation rate.¹¹⁵

The classic approach to weaning neonates from CMV is to extubate them from a rate of about 15 breaths/min to supplemental O₂ or nasal CPAP. Weaning strategies involve decreasing rate (except for patient-trigger ventilation) and positive inspiratory pressure to maintain the V_T at 4 mL/kg or greater. PSV may be combined with SIMV. Newer options include volume-assured pressure support and proportional-assist ventilation.¹¹⁶ In neonates, direct extubation from a low ventilator rate showed a trend toward greater successful extubation than did extubation after a period of endotracheal CPAP.¹¹⁷ Weaning from HFOV is performed by switching to CMV or by extubation after gradual reduction of mean airway pressure and amplitude to very low values.

Nasal CPAP, noninvasive positive-pressure ventilation, and methylxanthines are evidence-based treatments to facilitate weaning and extubation of preterm infants, but only the first two can be recommended for routine use. In contrast to adults, steroids are not recommended for prevention of postextubation stridor in pediatric or neonatal populations.¹¹⁸

In conclusion, neonates and children can be weaned much quicker than adults, and weaning protocols have proven to be of little or limited value in pediatric patients.

Common Techniques for Respiratory Support

The following invasive and noninvasive techniques are commonly used in infants and children to support spontaneous ventilation or to apply controlled mechanical ventilation:

1. CPAP during spontaneous breathing
2. PSV to assist spontaneous breathing
3. PCV or VCV
4. Intermittent mandatory ventilation (IMV) or SIMV, with or without PSV
5. High-frequency ventilation (HFV) with or without spontaneous breathing

Clinical experience with most of these modes in neonates and infants, although less in children, is distinctly different from that in adults. There is a long-standing neonatal experience of nasal CPAP and HFV. Conversely, assist modes only recently became available in the neonatal field because of difficulties in providing sufficiently sensitive trigger systems, especially for end-expiratory flow termination during PSV. This latter may be enhanced through the availability of neurally adjusted ventilatory assist (NAVA), which appears to lessen

asynchrony.¹¹⁹ All modes, with the exception of HFV, are also used during noninvasive ventilatory support, either by nasal or nasopharyngeal prongs or cannula, nasal or nasofacial masks, or the helmet.

Continuous Positive Airway Pressure in Neonates

In infants, CPAP is commonly applied via the nasal route with a short binasal cannula (nasal prongs), a single nasopharyngeal cannula, or a soft nasal mask. Short binasal prongs seem to be more effective than a single prong.¹²⁰ The feasibility of using a helmet as CPAP interface was documented recently.¹²¹ To apply CPAP, various methods, such as electronic feedback control, underwater seal, flow opposition, and flow opposition with fluidic flow reversal on expiration have been used. Each method imposes a different resistive load and there is ongoing discussion whether one is better than another^{122–125} in reducing work of breathing. For practical reasons, it is simpler to divide the commercially available devices into two types: continuous and variable flow systems.

Continuous flow systems adjust the CPAP level by either modulation of the resistance created by an valve in the expiratory circuit (the case for some ventilators with inbuilt CPAP modes) or by placing the expiratory tubing under water, the depth of the tubing allowing for blow-off at a certain CPAP level (the case for bubble CPAP systems). These systems require relatively high constant flow rates (often above 8 L/min) to ensure that flow during inspiration at least matches a patient's peak inspiratory flow. Simple high-flow systems can also be used for CPAP;⁵¹ their efficiency remains doubtful, and pressure variations are poorly controlled.¹²⁶

Variable flow systems maintain a set CPAP level by rapid adjustments in the flow rate. Flow modulation can result from either fast modulation of inspiratory flow or, with a special nose piece, by entrainment of gas at the inspiratory jets (Infant Flow system) or with a jet device at the nasal prongs (Benerviste system).¹²⁷ With the latter systems, CPAP is changed by increasing or decreasing flow. The pressure generated should be measured with an accurate manometer and linked to an alarm.

No clear evidence favors any particular nasal interface¹²⁰ or flow pattern.¹²⁸ A recent randomized controlled trial that compared various CPAP systems (bubble vs. infant flow) suggested certain advantages with a bubble CPAP system in terms of secondary outcome measures, such as a higher rate of successful extubation or shorter duration of CPAP in infants recovering from RDS.¹²⁹ A crossover study in premature infants showed better oxygenation with bubble CPAP when compared to ventilator-derived CPAP.¹²⁵ Such observations are intriguing and question the belief that modern expensive CPAP devices perform better than older, simple, and relatively cheap devices.

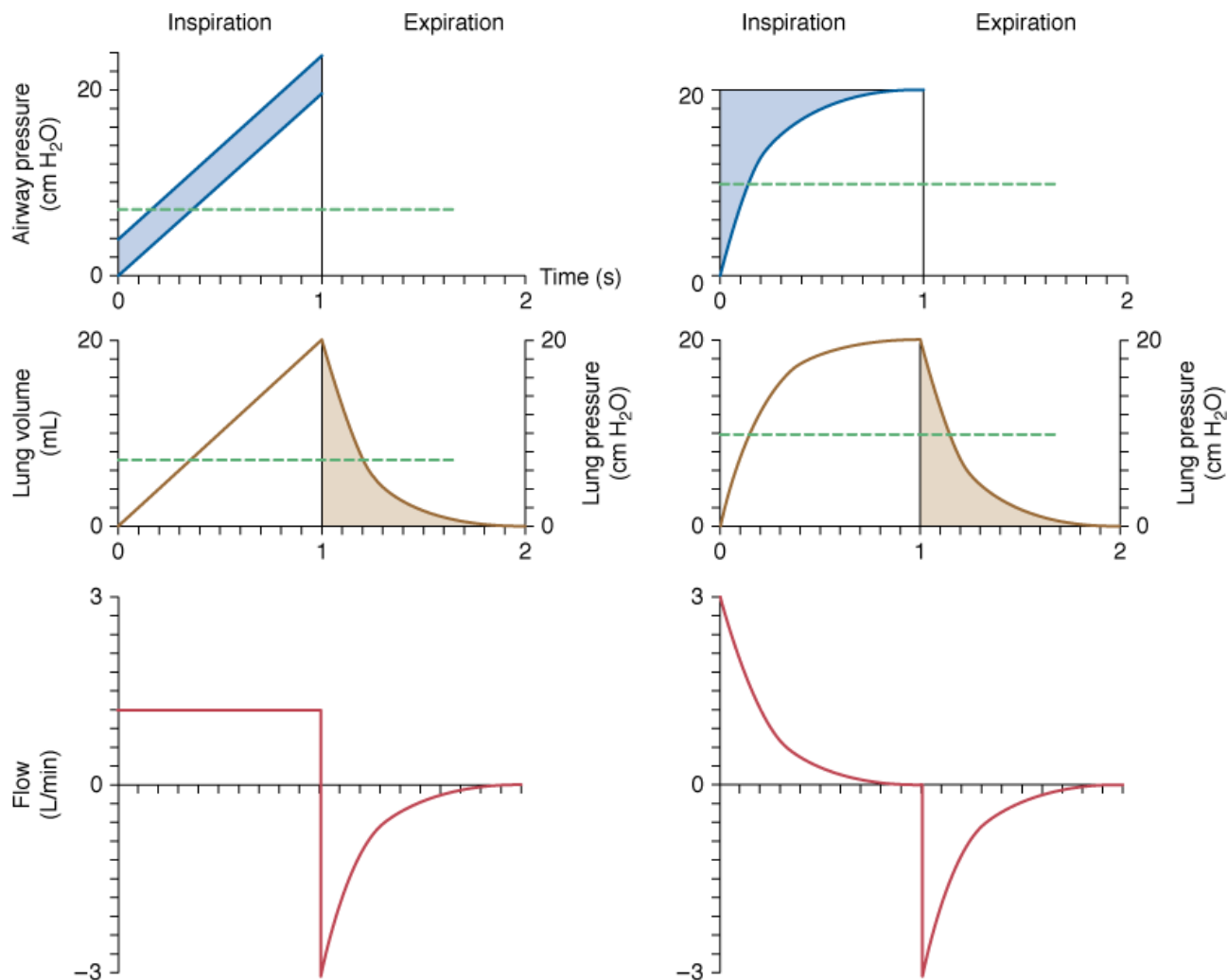
Nasal CPAP is now used as first-line support in the neonatal field, mainly for premature infants presenting with various forms of respiratory distress (classically, infant RDS). Many unanswered questions remain,¹³⁰ such as optimal pressure or flow to be used,¹³¹ whether early CPAP for RDS reduces mortality and morbidity (chronic lung disease remaining a significant problem in neonatal intensive care) as compared with intubation and mechanical ventilation (a large trial failed to give a definitive answer⁴⁹), and the lack of clear criteria to indicate when an infant is unresponsive to nasal CPAP.¹³²

Conventional Mechanical Ventilation in Neonates and Infants

Time-Cycled, Pressure-Limited Ventilation

Pressure-control techniques have been used in neonates for many years. Such neonatal *pressure controllers* deliver a continuous, constant flow during inspiration and expiration, and the inspiratory and expiratory pressure levels cycle at regular intervals (time-cycled, pressure-limited [TCPL] ventilation). In that inspiratory flow characteristics of TCPL ventilation differ from classic PCV, which modulates inspiratory flow in a decelerating pattern (Fig. 23-3), a classic neonatal ventilator that offers TCPL ventilation is nothing more than a simple and easy-to-operate *flow driver*. To achieve an inspiratory plateau pressure, relatively high flow rates are required (6 to 10 L/min), especially when short T_I values (0.3 to 0.4 second) are used to keep mean airway pressures low.

Figure 23-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Graphic representation of the equation of motion for a constant inspiratory flow pattern (*left*) and a constant inspiratory pressure pattern (*right*). The *dotted lines* indicate mean airway and lung pressures. The *shaded sections* represent equal geometric areas proportional to the pressure required to overcome flow resistance. The *unshaded sections* represent equal geometric areas proportional to the pressure required to overcome elastic recoil. Note that for the same tidal volume and mean inspiratory flow rate (i.e., tidal volume-to-inspiratory time, V_T/T_I), the constant flow pattern produces a higher peak airway pressure and lower mean airway pressure but the same peak lung pressure. (Reprinted, with permission, from Chatburn RL. Principles and practice of neonatal and pediatric mechanical ventilation. *Respir Care*. 1991;36:569–595.)

On several occasions, pressure-limited ventilators have failed to provide adequate alveolar ventilation in newborn babies; this can be a problem when T_I is too short or with insufficient inspiratory flows. A second and perhaps more important problem with TCPL ventilation is that delivered V_T varies from breath to breath, coupled with the fact that many flow drivers still do not offer measurements or display of delivered V_T . The same problems, however, hold true for any pressure-control mode.

A third problem with TCPL ventilation is the slow buildup of pressure and cycle to pause as the pressure target is achieved. There is argument as to whether this leads to slow intratidal recruitment of the airways during the insufflation phase, the quantity of which depends on the pressure delivered. This may create less shear force in the airways than does classic PCV with a decelerating flow pattern. Conversely, a potential disadvantage of TCPL ventilation is that as soon as the inspiratory pause time commences, unstable airways are prone to collapse as gas is redistributed to areas with longer time constants. Consequently, a situation arises whereby collapse is already commencing during the inspiratory phase. Conversely, if a fast regulation system is used (variable flow delivery in a pressure-controlled mode) that promptly delivers flow, tightly regulated by pressure, previously collapsed airways will continue to expand; others with a longer time constant experience a chance of opening throughout inspiration. This implies that there is no pause during the inspiratory period and that the regulation system is active even during the “no-flow” phase. Flow during this “no-flow” phase is seen only on high resolution but

adds to the efficacy of recruitment by providing extra time for lung inflation; this behavior is best explained by the power law of avalanches.¹³³

Pressure-Controlled Ventilation

Classic PCV has been introduced in neonatal ventilation only recently. It differs from TCPL ventilation in that inspiratory flow is variable (decelerating flow pattern). Peak inspiratory pressure can be reached early during inspiration. This leads to fast increases in circuit and airway pressures, which may help to overcome high airway resistance. For years, this was considered harmful, but no evidence supports this concern.

Volume-Controlled Ventilation

VCV in neonatology was abandoned in the early 1980s mainly because of difficulties in measuring very small V_T values under low-flow conditions. A second concern was the slow rise to peak pressures (with constant inspiratory flow), eventually causing unequal lung-volume distribution within the lung. A third concern was the frequent occurrence of an ETT leak (not using cuffed tubes), which can alter effective delivery of volume substantially. All these concerns are mere assumptions and not supported by clear scientific evidence. Because of technological advances and discussion of small V_T values for lung protection, there is growing interest in this mode again. A single-center, randomized, controlled trial has compared pressure-limited ventilation versus VCV, with identical V_T values in both arms, in newborns presenting with RDS.¹³⁴ There was a small benefit, in terms of ventilator duration and significantly less intraventricular hemorrhages and abnormal periventricular echo densities on ultrasound scans, in favor of VCV. The lower rate of cerebral complications with VCV may reflect better P_{CO_2} stability.

Volume-Targeted Ventilation

Recognition that volume rather than pressure causes ventilator-induced lung injury has led to the development of various techniques that combine features of pressure-limited and volume-limited ventilation in a mode, usually referred to as *volume-targeted ventilation* (VTV; i.e., volume guarantee, pressure-regulated volume control, or volume-assured pressure support). With this, delivery of a set V_T is “guaranteed” by adjusting inspiratory pressure according to changes in compliance, resistance, or respiratory drive. This usually requires a learning period over a number of breaths, during which dynamic compliance and respiratory system resistance are assessed by the ventilator. Small studies show only small breath-to-breath variability in V_T ,^{57,135} although there is concern about overshooting volume delivery over several breaths (learning period effect) if compliance or resistance change rapidly.³⁹ Along the same lines, inbuilt algorithms may cause an increase in airway pressure resulting in larger than targeted V_T over the next few breaths if a spontaneously breathing infant’s efforts have interrupted his or her expiration by diaphragmatic braking.¹³⁶ Therefore, the safety of these devices may need to be reconsidered.

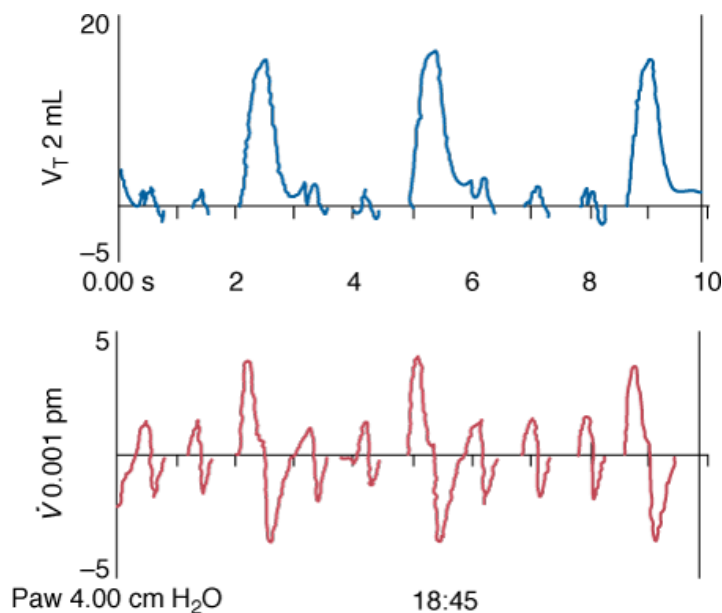
Recent meta-analytic data suggests that the use of VTV in premature infants with RDS might improve pulmonary and neurodevelopmental outcome.³⁷ Cross-sectional surveys document a rapid increase in the routine use of VTV,^{31,137,138} but with a considerable variation in VTV practice. However, despite the fact that some have provided practical advice on use of VTV in premature infants,¹³⁸ we are still far away from establishing best VTV practice.

Patient-Triggered Ventilation

Synchronized Intermittent Mandatory Ventilation

Asynchrony may cause poor gas exchange and air trapping, leading to pneumothorax, hemodynamic instability, changes in cerebral blood flow, and intracranial hemorrhage in newborns.¹³⁹ Synchronization of spontaneous inspiration with the ventilator has been attempted, initially with SIMV, which conducts a search for a spontaneous effort within a predefined time frame (Fig. 23-4). Although SIMV was thought to be superior to IMV, no evidence supports this suspicion.¹⁴⁰

Figure 23-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Volume (*above*) and flow (*below*) waveforms in a patient receiving synchronized intermittent mandatory ventilation (SIMV). The SIMV breaths occur at somewhat irregular intervals because of the timing windows, during which the machine waits to detect spontaneous effort by the patient. *Paw*, airway pressure; *V_I*, flow; *V_T*, tidal volume. (Reprinted, with permission, from Donn S. Invasive and noninvasive neonatal mechanical ventilation. *Respir Care*. 2003;48:426–439.)

Assist-Control, Intermittent Mandatory, and Pressure-Controlled Ventilation

Assist-control ventilation, IMV, and PCV were developed to assist all spontaneous breaths that exceed a trigger threshold; in the case of apnea or bradypnea, a mechanical breath is provided by the machine according the rate set. The inspiratory time is set by the operator. Asynchrony occurs when the machine's T_I exceeds the infant's T_I .

Pressure-Support Ventilation

It is possible to terminate a machine breath based on a decline in inspiratory flow to below a certain percentage of peak inspiratory flow, as with PSV. Algorithms such as this help to synchronize patient effort with ventilator assistance.¹⁴¹ PSV increases inspiratory pressure to a preset level and is intended to decrease the work of breathing. Flow delivery is variable and proportional to patient effort. Initially, PSV was viewed mainly as a weaning mode, but subsequently, it was used more widely in the adult field.¹⁴² Since the development of sensitive flow triggers and variable inspiratory flow termination, PSV can be used in neonates.¹⁴³

SIMV, assist-control ventilation, IMV, pressure-control ventilation, and PSV can be grouped under the generic term of *patient-triggered ventilation*. In neonates, and especially in preterm infants, the trigger needs to sense any weak spontaneous breathing effort. It also must minimize artifacts that may result from other sources: heartbeat-induced variations in intrathoracic pressure or variations in pressure oscillations induced by rainout in the ventilator tubing. Various trigger signals have been used: flow or pressure triggers, abdominal or thoracic impedance methods, and detection of abdominal motion. Flow triggering is more sensitive than pressure triggering for neonatal ventilation.¹⁴⁴ Two major sources of patient-ventilator dyssynchrony are (a) inappropriate long delays in inspiratory triggering (preterm infants with a short T_I may have completed much of the inspiratory phase before ventilator assistance commences) and (b) inappropriately selected inspiratory flow termination (“cycling off” or “flow termination” setting given as a percentage of maximum inspiratory flow; if set too low, inflation times will be too long). With this, the flow at which the ventilator cycles to exhalation does not coincide with the end of patient T_I .

Neurally Adjusted Ventilatory Assist

NAVA offers a new approach to patient-triggered ventilation. This mode provides ventilator support triggered by, and proportional to, the electrical activity of the diaphragm (see [Chapter 13](#)). Experience with NAVA in the neonatal field is sparse. In a small series of premature infants, better patient–ventilator synchronization (more appropriate cycling off) was observed as compared to PSV.¹⁴⁵ This was the case even with large air leaks, suggesting that NAVA may improve ventilator synchrony during noninvasive ventilation through nasal prongs or masks. Whether this will decrease the need for intubation in infants with RDS is unknown. In pediatric patients, synchrony has undergone less evaluation, although neural triggering has been shown to precede pneumatic triggering in more than two-thirds of breaths during NAVA.¹¹⁹

Conventional Mechanical Ventilation in Children

In pediatric intensive care units, conventional mechanical ventilation is used in a similar fashion to adult intensive care units. With today's ventilators, which offer various control (pressure or volume control) or assist (mainly PSV) modes, trigger systems (flow or pressure), and cycle criteria (time or flow based), even small children can be satisfactorily assisted. In the pediatric setting, PCV and PSV are most commonly used,² although volume control or VTV are sometimes preferred and useful for assuring stable P_{CO_2} values in patients with pulmonary hypertension (reactive pulmonary vasculature) or increased intracranial pressure. Otherwise, there is no proven benefit for these latter modes in children.

High-Frequency Ventilation in Neonates and Infants

HFV is defined by a frequency that greatly exceeds the normal respiratory rate and a V_T that approximates anatomic dead space (see [Chapter 19](#)). There are three major types of HFV:

1. *High-frequency positive-pressure ventilation* (rate: 60 to 150 Breaths/min)
2. *High-frequency jet ventilation* (HFJV; rate: 100 to 600 Breaths/min)
3. HFOV (rate: 180 to 1500 oscillations/min or 3 to 25 Hz)

HFOV is the most commonly used method in pediatric and neonatal intensive care units. High-frequency positive-pressure ventilation and HFJV promote gas exchange in a conventional way with V_T greater than dead-space volume. Expiration is passive. In contrast, HFOV uses V_T values that are less than dead space, and expiration is active.

HFJV is used mainly during (adult) laryngeal surgery; few randomized, controlled trials exist in neonates, with conflicting results.^{146,147} Concerns have been raised about prolonged HFJV not only in neonates but also in adults regarding airway damage, ranging from focal necrosis to complete airway obstruction with mucus and severe necrotizing tracheobronchitis.^{148,149} Increased risk of adverse cerebral outcome has been reported.¹⁵⁰ Studies have not used consistent criteria for assessment, however, and HFJV systems have varied widely.

All these methods are highly effective in eliminating CO_2 , but the effect on oxygenation is less uniform. This is one reason that these modes (especially HFOV) have failed to sustain their initial attraction. Within the context of ventilator-induced lung injury and lung-protective strategies, HFV could be viewed as the optimal protective mode. By providing very small V_T values, it is possible to ventilate the lung within a safe zone of the pressure–volume curve of the respiratory system.¹⁵¹ Side effects, such as from respiratory acidosis (permissive hypercapnia), are not associated with HFOV, and spontaneous ventilation can be often maintained, at least in neonates and small children. Thus, use of sedation is decreased, and muscle relaxants are avoided. In larger patients, inspiratory flow demands are higher; thus, spontaneous breathing is not managed as easily, and heavy sedation and/or paralysis may be required.

Indications and Timing for High-Frequency Oscillation Ventilation in Neonates and Children

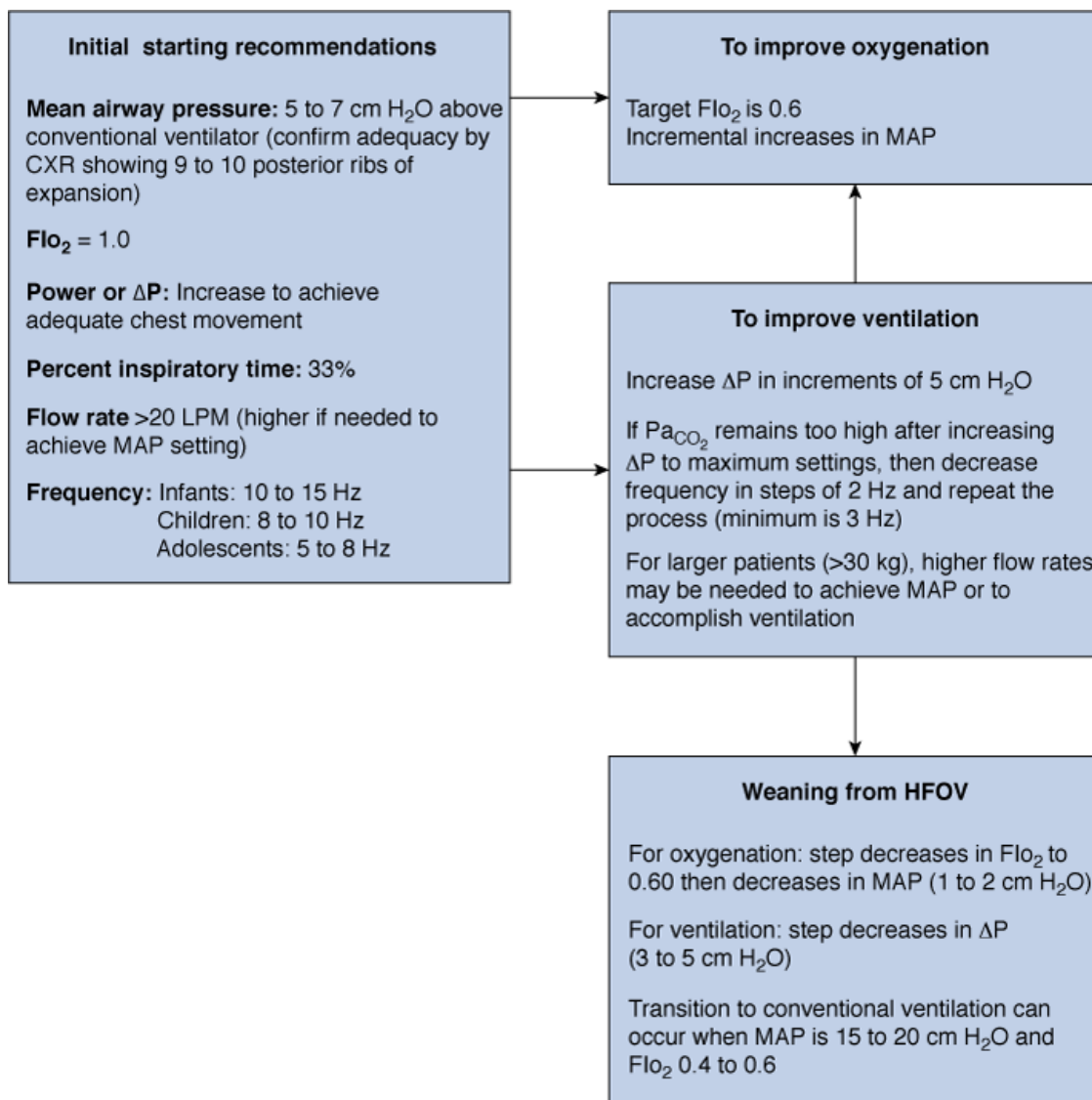
HFOV is still used mainly as rescue therapy despite clear indicators from experimental and clinical experience that it is most beneficial when initiated before major lung injury has developed^{152–155} because no ventilator strategy can repair preexisting lung injury.^{156,157}

Classic indications for HFOV in neonatal patients include:

1. *RDS* characterized by surfactant deficiency, high chest wall compliance, and low functional residual capacity. This is the constellation for easy and efficient lung recruitment, at least with early disease.^{153,156} Conceptually, the success of HFOV depends on its ability to recruit lung volume, which is not always easy late in disease when substantial ventilator-induced damage is superimposed on preexisting injury. Most randomized controlled trials, comparing HFOV to *CMV* in infants with *RDS*, produced controversial outcome data.^{28,59,158} This has been partially attributed to flaws in study design and noncompliance with study protocols.²⁷ A recent small randomized controlled trial (HFOV vs. *CMV*) in a well-defined patient group (minimizing heterogeneity) and employing comparable ventilator strategies in both arms, showed a benefit⁶⁰ in favor of HFOV. This may revitalize the debate on the use of HFOV in neonates with *RDS*.
2. *CDH*, usually associated with alveolar and pulmonary vascular hypoplasia,¹⁵⁹ resulting in increased risk of ventilator-induced lung injury¹⁶⁰ and presenting with *PPHN* requiring aggressive P_{CO_2} control. Lung recruitment should be applied with great caution because of the difficulty in estimating the degree of lung hypoplasia.
3. *Air-leak syndrome*, such as pneumothorax or interstitial emphysema. HFOV can achieve early improvement and resolution of air leak in patients with interstitial emphysema, enabling better gas exchange and lower airway pressures than with *CMV*.¹⁶¹
4. *PPHN*. Although *iNO* is the classic treatment for *PPHN*, HFOV can enhance delivery of *iNO* by achieving adequate lung recruitment¹⁶² and enable easier CO_2 control and correction of respiratory acidosis.

The classic indication for HFOV in pediatric patients is *diffuse alveolar disease (primary or secondary ARDS)*^{152,163} as rescue therapy to improve oxygenation. The only pediatric randomized controlled trial¹⁵² did not show any difference in survival or duration of ventilation duration, although duration of supplemental O_2 was shorter with HFOV than with *CMV*. Recommended strategies for use of HFOV (Fig. 23-5) are empirical and derive mainly from the neonatal field and experience with HFOV in adult patients.¹⁶⁴

Figure 23-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Recommendations for instituting high-frequency oscillatory ventilation (HFOV) and the transition to conventional ventilation during weaning. CXR, chest X-ray; LPM, liters per minute; MAP, mean airway pressure. (Courtesy Martha Curley, Children's Hospital Boston, Harvard Medical School. See also ref. 89.)

Unique Neonatal and Pediatric Machines and Interfaces

Neonatal and Pediatric Ventilators

Conventional pediatric and/or neonatal ventilators can be divided in three groups: (a) neonatal ventilators, (b) pediatric ventilators (which generally are identical to adult ventilators), and (c) all-age-group ventilators (from neonates to adults). Traditionally, a neonatal ventilator was used for neonates or toddlers up to a body weight of 10 kg. Neonatal ventilators were mainly constant-flow generators that build up pressure and cycle to pause as the pressure target is achieved (TCPL ventilators). This type of ventilators are limited to neonates and infants, because inspiratory flow does not allow a buildup of sufficient airway pressure in bigger patients and such ventilators are not able to adapt to the high inspiratory peak flows of bigger patients with spontaneous breathing activity. Today, some neonatal ventilators, mainly all-age-group ventilators, also offer classic pressure-control and volume-control modes.

The following are the characteristics of a good neonatal ventilator: (a) small apparatus dead space; (b) stiff but flexible circuit tubing; (c) reliable tidal-volume measurement device (hot-wire or pressure-differential flow meter or fixed or variable orifice device) placed between the Y piece and ETT; (d) highly sensitive trigger device (flow triggering preferred to pressure triggering at least in premature infants,¹⁴⁴ which provides adjustable inspiratory flow triggering and in PSV modes cycles off criteria ranging from 5% to 25% or more of peak

inspiratory flow); (e) built-in leak compensation, which can be turned off if there is a large leak around the ETT (this is important with PSV to enable inspiratory flow termination so as to avoid excessive T_I in relation to patient need); and (f) airway humidification with cascade humidifiers and heated-wire ventilator circuitry.¹⁶⁵

The new-generation all-patient units offer all these neonatal requirements and perform as well as those designed specifically for neonatal and pediatric patients.¹⁶⁶ Detailed specifications on “only neonatal” and “all-patient” ventilators can be found in Tables 23-3 and 23-4 in the second edition of this book.

High-Frequency Ventilators

Important differences in performance characteristics of devices for HFV have been shown repeatedly.^{167–170} Thus, it is not possible to compare settings from one device to another. Three major systems are used: HFOV, HFJV, and HFV with a flow-interrupter (HFFI) device.

High-Frequency Oscillation Ventilation

HFOV is characterized by use of mean airway pressure (continuous distending pressure), on top of which square-wave or sinusoidal pressure swings are added by means of a piston, diaphragm, or bidirectional high-velocity flow in the expiratory limb. This leads to a more or less active aspiration during the “expiratory” phase of the oscillatory cycle. The best studied and most widely applied HFOV system is the SensorMedics 3100 high-frequency oscillator (VIASYS Healthcare, Palm Springs, CA). This exists in two versions: the 3100A for neonatal use and the 3100B for pediatric and adult use. The Humming V (Metran Medical Instruments, Saitama, Japan) exhibits similar performance to the SensorMedics 3100A in vitro and bench studies and also can be switched to CMV. Other neonatal ventilators combine conventional and HFOV. There is tendency to use the term HFOV for various forms of HFV, but to qualify as an oscillator, a device has to generate a waveform that has both positive and negative pressure deflection. Each system has a slightly different design (piston or diaphragm, Venturi-valve or jet injector) for creating biphasic pressure waveforms (high frequency oscillations). Therefore, the oscillation power of each machine is slightly different from the other, which explains their various ability to support children only to a certain weight limit.

A new development in many neonatal oscillators (Fabian, Leoni-plus, and VN500) is the ability to servo-control the delivered V_T (measured as exhaled or inhaled tidal volume at the airway opening) by means of HFOV+Volume Targeted Ventilation (volume guarantee) option. This allows the user to preset a V_T during HFOV. Whether this new mode offers any advantage is unknown; using target V_T ventilation during HFOV has no sound background and it goes against the principle of maintaining a mean airway pressure for lung-volume optimization and superimposing the lowest amplitude needed for acceptable CO_2 washout. We still do not know which V_T is appropriate in the various situations, given that V_T during classic HFOV shows large variations (0.5 to 2.5 mL/kg). V_T depends on the oscillation frequency and there is no consensus on the optimal frequency in various settings. Given these unknowns, HFOV with the VTV option should not be used until more data on its safety are available. Making extrapolations based on protective V_T during conventional ventilation to V_T during HFOV is inappropriate because of completely different gas-exchange mechanisms (convection vs. molecular diffusion) between the two modes.

High-Frequency Jet Ventilation

HFJV is characterized by the delivery of small, high-velocity breaths at fast rates coupled with passive exhalation. HFJV allows the application of relatively low airway pressure to maintain reasonable oxygenation. Pulses of gas are delivered at high velocity through an orifice at frequencies of 10 to 100 Hz. The orifice may be in a T-piece connected to a conventional ETT or a narrow tube incorporated in the wall of a special ETT or at end of a fine-bore catheter placed in the trachea. Available HFJVs for pediatric use include the Acutronic Monsoon Deluxe Jet Ventilator (Acutronic Medical Systems, Hirzel, Switzerland), and for pediatric and neonatal use the Life Pulse High Frequency Ventilator (Bunnell, Salt Lake City, UT). In the pediatric field, indications for HFJV are available in the operating room during endolaryngotracheal surgery.¹⁷¹ In the neonatal field, HFJV has been proposed as rescue treatment for severe pulmonary dysfunction in infants with RDS, CDH, and meconium aspiration syndrome.^{147,172–174} Although promising results have been reported with HFJV as early or late rescue¹⁷² treatment in infant RDS, benefit in terms of pulmonary or neurodevelopmental outcome has not been documented.¹⁷⁵

High-Frequency Flow-Interrupter

For HFFI (basically a [CMV](#) mode) gas flow from a high-pressure source is interrupted at a high rate. Expiration is passive, but additional negative pressure generated by a Venturi effect on the expiratory side may enhance lung emptying. The Infrasonics Infant Star 950 (Nellcor Puritan-Bennett, Inc., Pleasanton, PA) is an example. HFFI devices are available only for neonatal ventilation, and little data exist on their clinical use.^{176,177}

Interfaces

Much progress has been made in improving interfaces. For the invasive airway management, improvements in ETT design have changed the traditional teaching to avoid use of cuffed ETTs in infants and children younger than 8 years of age because of increased risk of subglottic injury. This dogma had inhibited research on this topic despite major technological advances with the development of low-pressure, high-volume cuffs and new material. Recent studies show that cuffed tubes can be safely used in children of all ages, provided the correct size is used and cuff pressure is continuously monitored, without an increase in postextubation stridor.^{178,179} Advantages include a lower reintubation rate, and control of leakage causing less autotriggering and more stable ventilation. Pediatric advanced life support recommendations have also changed and now suggest the use of cuffed tubes of appropriate size for emergency tracheal intubations in children.¹⁸⁰

For noninvasive ventilation, the fit and comfort of an interface together with patient-ventilator synchrony are the most important factors for success. There have been significant improvements in mask designs and in plastic materials used for mask seal interfaces (air-cushion seal, gel interfaces). The availability of bilevel-pressure masks for pediatric patients, however, is still very limited, and one has often to rely on petite versions of adult masks.¹⁸¹ Aside from the known complications related to nasal mask ventilation (skin irritation or breakdown, headache, eye irritation, gastric distension, leakage), certain problems and complications are unique or more pronounced in children. Leakage can simulate patient effort, causing some bilevel generators to autotrigger when set in spontaneous or timed mode. In these circumstances, the ventilator can be set in a timed or control mode at a rate that overrides the patient's respiratory drive during sleep. The pressure on facial structures from a nasal mask is associated with flattening of facial structures, especially the midface, leading to malocclusion with retrusion of the maxillary ridge.^{182,183} The choice of the interface also influences the type of ventilator used. Nonvented masks are designed for use with more sophisticated intensive care unit-type ventilators (requiring an active exhaust valve), whereas vented masks are used for the more traditional flow-cycled bilevel-pressure home ventilator (not requiring an active exhaust valve).

Monitoring of Mechanical Ventilation: Specific Pediatric and Neonatal Considerations

Several specific pediatric and neonatal issues about respiratory monitoring must be noted. First, ventilator-circuit dead space should be kept as small as possible when monitoring devices are added. Second, flow measurements must be made as close as possible to the airway opening to get true delivered V_T .^{184,185} Third, with standard practice of using uncuffed ETTs air leakages around the ETT are observed in approximately 70% of all ventilated neonatal infants, rendering tidal volume monitoring less reliable.¹⁸⁶ Fourth, little is known about the reliability of end-tidal CO_2 measurements in the presence of ETT leakage (>30%).¹⁸⁷ Fifth, accurate measurements of exhaled CO_2 in pediatric and especially neonatal patients is challenging because of limitations in size of the analyzer chamber (to reduce instrumental dead space), the amount of flow that must be suctioned by sidestream devices, and the need for fast response times because of rapid respiratory rates. Therefore, small dead-space volume mainstream sensors, which have a faster response time (than side-stream devices) and do not need suction flow, are preferred.¹⁸⁸ Recently, a microstream capnograph (suction flow of only 30 mL/min) with a miniaturized sample chamber was developed, thus, improving the measurement accuracy of end-tidal CO_2 in intubated infants.^{189,190} Volumetric capnography, measuring CO_2 elimination per breath (V_{CO_2}), may assist with optimal PEEP titration: atelectatic or overdistended areas reduce alveolar ventilation and therefore CO_2 elimination.^{191,192} The concept is used by companies that offer inbuilt volumetric capnography (e.g., “the open lung tool” on the Servo-I ventilator by Maquet, Solna, Sweden). V_{CO_2} monitoring has also been proposed for guiding weaning but the concept has been questioned.¹⁹³

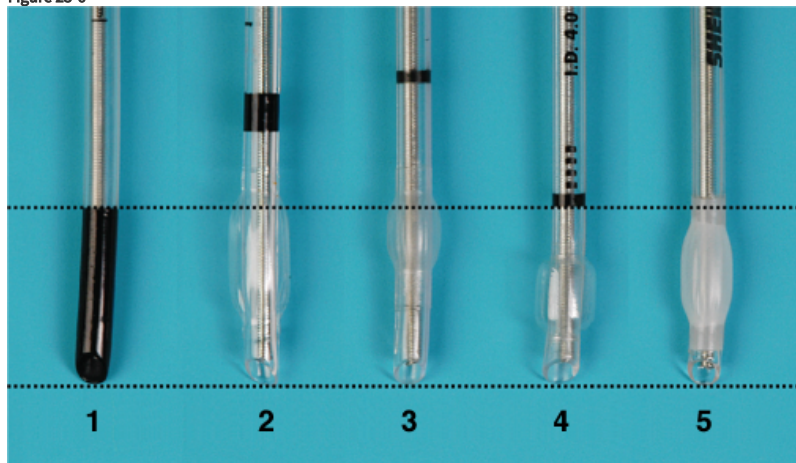
Complications of Mechanical Ventilation

In addition to barotrauma (e.g., pneumothorax and interstitial emphysema), three other complications include postextubation stridor, development of chronic lung disease or BPD (classic in the preterm infant), and ventilator-associated pneumonia.

Postextubation Stridor

Acquired subglottic stenosis became a well-recognized problem with use of long-term endotracheal intubation. Serious tube trauma and postextubation stridor have decreased markedly with improvements in material and tube design and greater care in choosing appropriately sized tubes. Tube complications usually arise from incorrect size, traumatic or multiple intubations, up-and-down movements of the tube, and inadequate analgesia and sedation, causing intubated infants to struggle. Poor design can result in a cuff being positioned too high within the larynx (Fig. 23-6), causing severe trauma.¹⁹⁴ These observations have led to new design for ETTs.^{195,196} With newly designed cuffed-ETTs (cuff pressure held ≤ 20 cm H₂O), side effects, such as an increased rate of postextubation stridor, were not observed in large observational studies^{178,197–199} and in one randomized controlled trial.¹⁷⁹ Specifically designed cuffed ETTs, such as the Microcuff PET tubes, have proven safe in neonates of birthweight greater than 3 kg and in children,²⁰⁰ and no real arguments remain against their use.²⁰¹ Other risk factors of postextubation stridor include duration of intubation, gastroesophageal reflux, and tracheobronchial infection. Checking for a cuff leak is of little help in predicting postextubation problems in young children.¹⁰⁴

Figure 23-6



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Positioning of the cuff on various endotracheal tubes: (1) uncuffed tracheal tube, inside diameter 4.5 mm; (2) Rueschelit Super Safety Clear, inside diameter 4 mm; (3) Mallinkrodt Hi-Contour P, inside diameter 4 mm; (4) Microcuff PET, inside diameter 4 mm; and (5) Sheridan CF, inside diameter 4 mm. Note that only tube 4 has a cuff of appropriate size and positioning. (Courtesy Markus Weiss, Pediatric Anesthesia, Children's Hospital of Zurich, Switzerland.)

Most injuries causing postextubation stridor are superficial and include nonspecific changes such as laryngeal edema, granulation tissue, and ulcerations. Stridor usually resolves with medical therapy. Inhaled [epinephrine](#) is successful in treating subglottic edema. If reintubation is required, a smaller ETT should be used to prevent additional trauma. The nasal route is preferred to optimize tube stabilization and to minimize the tube shifting with head movement. The child should be weaned rapidly from the ventilator to humidified O₂ delivered via a light T tube. These measures prevent infants from pushing out the tube with their tongue and allow for better head movements, thereby minimizing risk of further trauma. Alternatively, the child must be heavily sedated and eventually receive neuromuscular blockade. In most cases of simple subglottic edema, extubation is successful after a rest period of approximately 2 to 3 days with or without a short course of [dexamethasone](#) (1 to 2 mg/kg/day in divided doses). It is reasonable to administer [dexamethasone](#) to patients at risk of stridor for 24 hours before and 24 hours after extubation. Such prophylactic steroids may reduce postextubation stridor and the rate of reintubation.²⁰²

Bronchopulmonary Dysplasia in the Preterm Baby

The pathogenesis of BPD is multifactorial.^{203,204} The principal risk factors are lung immaturity, barotrauma, volutrauma, O₂ toxicity, prenatal and nosocomial infections, and increased pulmonary blood flow secondary to a patent ductus arteriosus.^{205,206}

Changes in neonatal care of very-low-birthweight infants, including antenatal corticosteroids, postnatal surfactant, and modified respiratory support, have improved survival and have changed the clinical picture of BPD. “Old” BPD described above is a consequence of ventilator-induced lung injury, whereas “new BPD” is mainly a developmental disorder in which the immature lung fails to reach its full structural complexity.^{204,207} Improved survival of very immature infants has led to more infants with this disorder. Changes in clinical practice have improved the clinical course and outcomes for infants with BPD over the past decade, but the overall incidence has not changed²⁰⁸ and there are large variations among centers.^{153,209–214} This wide range reflects the heterogeneity of study populations, management practices, and disease definitions. The initial description of BPD by Northway et al⁵² and Bancalari et al²¹⁵ was based on O₂ requirements at 28 days of age. As newborns of lower gestational age survived, Shennan²¹⁶ later proposed the term *chronic lung disease*. The diagnosis of BPD is currently based on the need for supplemental oxygen for at least 28 days after birth, and its severity is graded according to the respiratory support required at 36 postmenstrual weeks.²¹⁷

Management of BPD includes minimizing ventilator duration and avoiding high O₂ concentrations. Fluid restriction, diuretics, and bronchodilators are helpful but do not alter disease course. Pulmonary vasodilators are not beneficial for BPD-associated pulmonary hypertension.²¹⁸ Dexamethasone can accelerate ventilator weaning and shorten the period of O₂ supplementation but is associated with worse neurologic outcome (increased leukomalacia, impaired brain growth) and more frequent severe gastrointestinal complications.^{219,220} Dexamethasone therapy, if given at all, should be brief²²¹ and early, but initiated only beyond the first week of life.²¹⁹ Hydrocortisone has been recommended in place of dexamethasone because the latter may impair neurodevelopment. The efficacy of hydrocortisone, however, is questionable.²²² Inhaled steroids have some efficacy, but only in ventilated patients.²²³ Despite several attempts for severity grading and various approaches for treatment and prevention, long-term prognosis remains uncertain.^{224,225}

Ventilator-Associated Pneumonia

Ventilator-associated pneumonia (VAP) is the second most common nosocomial infection in the pediatric intensive care unit.^{226–228} VAP occurs in approximately 5% of ventilated children, and nearly 20% of affected children die. *Pseudomonas aeruginosa*, enteric gram-negative bacilli, and *Staphylococcus aureus* are isolated most commonly from endotracheal aspirates, particularly with VAP of late onset.²²⁹ Specific risk factors associated with pediatric VAP include the presence of a genetic syndrome, tracheal reintubation, transport out of the pediatric intensive care unit, subglottic or tracheal stenosis, trauma, and tracheostomy.²³⁰ VAP is associated with increased pediatric intensive care unit length of stay, ventilator days, and mortality rates. Adherence to a VAP-prevention bundle decreases VAP rate in pediatric patients.²³¹ For diagnostic purposes, protected bronchial brushing (sensitivity 72%, specificity of 88%) offers an alternative to poorly supported bronchoscopic bronchoalveolar lavage.²³²

Important Unknowns

Mechanical ventilation has increased survival of acutely ill neonates and children. A better understanding of pathophysiology has led to newer ventilator strategies. Many of these strategies, however, have not been tested in the pediatric setting, but simply adapted from adult experience. Recent research suggests that comparable ventilator settings are more injurious in the adult than in the infant lung.²³³ This latter is supported by conflicting results from observational cross-sectional studies investigating the relationship between V_T and outcome.^{30,32}

Approaches to optimizing ventilation are based on bedside assessment of respiratory mechanics and blood-gas response. Unfortunately, no characteristic of the pressure–volume curve can predict end-expiratory atelectasis, overstretching, or optimal airway pressure.^{234,235} Despite improved understanding of pediatric diseases and mechanical ventilation, we still do not know the best settings for an individual child. To date, evidence is confined to bad ventilator settings.

Noninvasive ventilation has proven safe and feasible in pediatric settings and can help to avoid intubation, although we do not know whether to avoid intubation is a wise decision in some conditions. This latter is true for the pediatric and neonatal field.^{49,236,237} Ventilator discontinuation is believed to decrease several complications, but proof is limited.

The Future

Respiratory monitoring has been used to decrease complications and improve patient–ventilator interactions. We need to develop approaches whereby this knowledge can be used to improve patient outcome. Tools that provide continuous recordings of pulmonary function have yet to be incorporated into ventilator equipment. For example, the continuous monitoring of functional residual capacity might improve ventilator strategies and enhance assessment of lung recruitment. Newer imaging techniques, such as bedside ultrasonography, electrical impedance tomography,²³⁸ or optoelectronic plethysmography,²³⁹ may help to monitor regional ventilation.^{240–242} In the end, such techniques will need to prove cost-effectiveness and/or improve outcome. Furthermore, we need a better description of what constitutes “standard” care.

Summary and Conclusion

Mechanical ventilation of infants or small children requires fundamental knowledge of the anatomic and functional characteristics of the respiratory system in a growing child. Clinicians also require good knowledge of specific pediatric pathologies, which sometimes results in a different approach to ventilation than in an adult patient. Nevertheless, adult experience has influenced many strategies applied in pediatric and even neonatal settings because it remains difficult to do well-controlled trials of mechanical ventilation in the pediatric intensive care unit.

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